

Optimising Asimov Bibliography with Integer Linear Programming

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16 August 2026

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1 Introduction

When people decide to read all the bibliography of one author for whatever reason, the natural question to ask oneself is: Which books should I buy to read them all? In principle this question can be trivially solved –i.e. just buy them all– however, regarding Asimov’s science fiction, this hardly is the case. Yes, one could in theory buy all of his bibliography but this not optimal. The main reason being redundancy. There is a lot of public literature regarding the extension of Asimov’s bibliography, specifically science fiction, here, a general summary is presented which motivates the problem to be solved.

Through the years, Asimov wrote a series of novels and short fiction stories. Part of this work was specifically written for science fiction magazines which means his work was originally printed in independent magazine volumes and even independent magazines. To address this issue, many editorials began to print collections of stories. The most famous being *I, Robot* or *The Foundation*, first published in 1950. Through the years of his writing many more short stories came and with them, many more collections.

This solves the issue of having to buy tons of magazines just to read five pages of an Asimov story however it is not enough. The collections present a major flaw: they repeat themselves. One might think that, if Asimov wrote 40 stories, then one collection would have the first 10, a second the next 10 and so on, but this is not the case. One collection has some stories, another collection has some other stories, which might overlap.

So the question is: which collections must one read to minimize overlapping. One example of this is the collection of *I, Robot* which is entirely included in the collection *The Complete Robot*.¹

This is an instance of the well known Set Covering Problem which is NP-Complete, but in these magnitude, the optimal solution can still be found and so, solve the question: which collections must I buy minimizing overlapping. However, due to the nature of our problem, there still will be a modification. One consequence of printing new collections is that people have to buy them, and in order to maximize the number of buyers many collections include exclusive stories which can only be found in said collection. For the Set Covering Problem solver this means that even if two collections overlap in all their stories except for one, the optimal solution will have to include both collections so the exclusive stories are counted.

Because our love for our authors is limited by our money, we have to make the hard decision of leaving some stories unread. So, the question we want to answer is: Which collections should one buy minimizing overlapping such that a certain threshold of money is not exceeded?

¹Actually *I, Robot* has some passages between stories narrated by one character that serve as a link from one story to the next one. However, those are not part of any short story and were written a posteriori, specifically for this collection.

2 Formal definition

Let $\Omega = \{s_1, \dots, s_n\}$ denote the set of all Asimov stories –i.e. short stories and novels–. Let $A \subseteq \Omega$ be a set of stories representing a collection and $\mathcal{C} = \{C_1 \dots C_m\}$ the set of all published collections. Note that because novels are also included, they are counted as stories inside singleton collections. For example, if s_k is a novel, then there is a collection containing solely s_k ($\exists C \in \mathcal{C}$ s.t. $C = \{s_k\}$).

Each story has a weight function $\omega : \Omega \rightarrow \mathbb{N}$ which gives the number of pages of a story s . Similarly for a collection $C \in \mathcal{C}$ its weight function is defined as the sum of the pages of all stories in C , that is:

$$\omega(C) = \sum_{s \in C} \omega(s) \quad (2.0.1)$$

A solution $S \subseteq \mathcal{C}$ is a set of collections. For a solution S , two weight functions are defined:

$$\omega^*(S) = \omega\left(\bigcup_{C \in S} C\right) \quad (2.0.2)$$

$$\omega(S) = \sum_{C \in S} \omega(C) \quad (2.0.3)$$

The difference between $\omega^*(S)$ and $\omega(S)$ is that the first gives the number of pages of distinct stories inside collections in S while the latter gives the total number of physical pages. For example, if S contains two collections C_1 and C_2 and both C_1, C_2 contain a story s , then ω will count the pages of s two times while ω^* only one.

With this notation, the optimization problem becomes

$$\begin{aligned} \min_{S \subseteq \mathcal{C}} \quad & \omega(S) \\ \text{s.t.} \quad & \omega^*(S) = \max_{S' \subseteq \mathcal{C}} \omega^*(S') \\ & \omega(S') \leq N \end{aligned} \quad (2.0.4)$$

where $N \in \mathbb{N}$ is the maximum number of pages.

Expressed like this, the problem can be thought as a Weighted Partial Set Cover Problem, however this name has already been used at (Bothe & Horváth, 2016) with a slightly different definition so it will not be used here.

2.1 Integer Linear Programming formulation

The problem defined at (2.0.4) can be formulated as a linear program. Let's take a look at the inner maximization problem.

$$\begin{aligned} \max_{\substack{x \in \{0,1\}^n \\ y \in \{0,1\}^m}} \quad & c'x \\ \text{s.t.} \quad & Ay \geq x \\ & c'Ay \leq N \end{aligned} \tag{2.1.1}$$

where $x_i = 1$ if story s_i is present in at least one of the chosen collections and 0 otherwise. $c_i = \omega(s_i)$ (the number of pages of s_i), y_j represents collection C_j being picked (1) or not (0). N is the maximum number of physical pages and $A \in \mathcal{M}_{n,m}(\{0,1\})$ is an incidence matrix expressing the inclusion relation between s_i and C_j , that is:

$$[A]_{ij} = \begin{cases} 1 & : s_i \in C_j \\ 0 & : s_i \notin C_j \end{cases}$$

Since y is an auxiliary variable semantically defined over x , there must exist some restrictions that enforce the link between x and y . We need, for all i :

$$x_i = 1 \Leftrightarrow \exists j \text{ s.t. } s_i \in C_j, y_j = 1 \tag{2.1.2}$$

which reads as: "A story s_i is present if and only if there exists a chosen collection that contains s_i ".

The implication from left to right is enforced by the first restriction $Ay \geq x$. The implication from right to left could be achieved by the restriction $mx \geq Ay$ but it is implicitly enforced by the maximization nature of the problem without any additional constraint. A proof is available in remark 1 of appendix A.

Adding the outer minimization problem

$$\begin{aligned} \min_{\substack{x \in \{0,1\}^n \\ y \in \{0,1\}^m}} \quad & c'Ay \\ \text{s.t.} \quad & Ay \geq x \\ & c'x = \max_{\substack{\tilde{x} \in \{0,1\}^n \\ \tilde{y} \in \{0,1\}^m}} c'\tilde{x} \\ & \text{s.t. } A\tilde{y} \geq \tilde{x} \\ & c'A\tilde{y} \leq N \end{aligned} \tag{2.1.3}$$

This problem's solutions also verify statement (2.1.2), but the reason is slightly more complex. A proof of it can be found in remark 2 of appendix A.

2.2 Buyer constraints

When put in practice a regular buyer would need more autonomy. Right now the *optimal* solution assumes the buyer hasn't got any collection, is able to buy any collection and wants to read all stories. This will most probably never be the case. For example *Have You Seen These?* was only published in 1974 as a special edition and probably will not be available in your local bookshop ever. So, to generalize even more the optimization these there new factors will be taken into account:

- (1) **Initial Collection Set:** The current collections a buyer already has. Represented as I .
- (2) **Goal Title Set:** The set of stories the buyer wants to own. Represented as G .
- (3) **Unavailable Collection Set:** The set of collections that cannot be bought. Represented as U .

The three constraints can easily be represented in the linear programming problem form.

(1) and (3) would be the constraints

$$y_j = 1 \text{ for all } j \in I$$

$$y_j = 0 \text{ for all } j \in U$$

To represent (2) the objective function of the inner maximization must be modified. Let M be a diagonal matrix such that

$$[M]_{ii} = \mathbb{I}_G(i) = \begin{cases} 1 & : i \in G \\ 0 & : i \notin G \end{cases}$$

then the objective function becomes

$$c'Mx$$

However it is not enough to add this constraints to problem (2.1.3) as the proof of the property (2.1.2) is no longer valid. Nevertheless for the inner maximization problem there is an easy workaround. Given an optimal solution that doesn't hold property (2.1.2) we can construct a second optimal solution that does hold the property. A proof of this is found in remark 3 of appendix A. Then the inner problem becomes:

$$\begin{aligned} \max_{\substack{x \in \{0,1\}^n \\ y \in \{0,1\}^m}} \quad & c'Mx \\ \text{s.t.} \quad & Ay \geq x \\ & c'Ay \leq N \\ & y_j = 1 \text{ for all } j \in I \\ & y_j = 0 \text{ for all } j \in U \end{aligned} \tag{2.2.1}$$

This effectively means that if for this problem we only care for the optimal objective value, which we do, then we know it must exist a solution that holds the property that has that optimal value, but we do not have to construct it.

Similarly if we just add the buyer constraints to the minimization problem we would obtain an optimal solution that not necessarily holds the property but, again, another optimal solution can be constructed that does hold it (see remark 4 of appendix A for a proof). The final problem then is:

$$\begin{aligned}
& \min_{\substack{x \in \{0,1\}^n \\ y \in \{0,1\}^m}} c' Ay \\
& \text{s.t.} \quad Ay \geq x \\
& \quad y_j = 1 \text{ for all } j \in I \\
& \quad y_j = 0 \text{ for all } j \in U \\
& \quad c' Mx = \max_{\substack{\tilde{x} \in \{0,1\}^n \\ \tilde{y} \in \{0,1\}^m}} c' M\tilde{x} \\
& \quad \text{s.t.} \quad A\tilde{y} \geq \tilde{x} \\
& \quad \quad c' A\tilde{y} \leq N \\
& \quad \quad \tilde{y}_j = 1 \text{ for all } j \in I \\
& \quad \quad \tilde{y}_j = 0 \text{ for all } j \in U
\end{aligned} \tag{2.2.2}$$

References

Bothe, S., & Horváth, T. (2016). The partial weighted set cover problem with applications to outlier detection and clustering. *LWDA*, 335–346.

A Proofs

Remark 1. *Constraint (2.1.2) is held for optimal solutions in problem (2.1.1)*

Proof. $\Rightarrow$ If $x_i = 1$ then because of the first constraint

$$1 = x_i \leq [Ay]_i = \sum_{j=1}^m a_{ij} \cdot y_j$$

$\Rightarrow \exists j$ s.t. $a_{ij} = 1, y_j = 1 \Leftrightarrow \exists j$ s.t. $s_i \in C_j, y_j = 1$.

$\Leftarrow$ Observe that constraint 2.1.2 must hold only for optimal solutions. If P is the feasible region, it is alright if some points $(x, y) \in P$ do not fulfill the constraint provided that optimal solutions do. We assume then, that (x^*, y^*) is an optimal solution.

By contradiction, if $\exists i$ s.t. $x_i^* = 0$ and $\exists j$ s.t. $s_i \in C_j, y_j^* = 1$ then we can construct another solution $(\tilde{x}^*, y^*)$ where $\tilde{x}^* = x^* + e_i$, (we take x^* and we set the i -th entry to 1).

We can see that $(\tilde{x}^*, y^*) \in P$ because $[Ay^*]_i \geq 1 = \tilde{x}_i^*$.

However, since all entries of c are strictly positive

$$c' \tilde{x}^* = c' (x^* + e_i) = c' x^* + c_i > c' x^*$$

which means that (x^*, y^*) is not an optimal solution and that is a contradiction. $\square$

Remark 2. *Constraint (2.1.2) is held in problem (2.1.3)*

Proof. $\Rightarrow$ Identical to $\Rightarrow$ in remark 1

$\Leftarrow$ Let's first observe that $mx \geq Ay$ is equivalent to the right to left implication in (2.1.2). This is because $0 \leq [Ay]_i \leq m$ (as A is a binary matrix and y has m components) and if $[Ay]_i > 0$ then it must hold that $x_i > 0$.

Let P_1 be the feasible region of the inner maximization problem and P_2 the feasible region of the whole problem.

Let $(x^*, y^*) \in P_2$ be an optimal solution and $(\tilde{x}, \tilde{y}) \in P_2$ such that $m\tilde{x} \not\geq A\tilde{y}$

We know that because $(\tilde{x}, \tilde{y}) \in P_2$ then $c'\tilde{x} = c'x^* = z$ where z is the optimal value of the maximization problem. Since $m\tilde{x} \not\geq A\tilde{y}$ then $\exists i$ s.t. $[A\tilde{y}]_i > 0$ and $\tilde{x}_i = 0$.

We will now show that $c'A\tilde{y} > c'Ay^*$. By contradiction, if $c'A\tilde{y} \leq c'Ay^*$ we construct $\bar{x} = \tilde{x} + e_i$ and since $c'A\tilde{y} \leq c'Ay^* \leq N$ then $(\bar{x}, \tilde{y}) \in P_1$.

But

$$c'\bar{x} = c'(\tilde{x} + e_i) = c'x^* + c_i > c'x^* = z$$

which means that z is not the optimal value of the inner maximization problem, which is a contradiction.

So, since $c'A\tilde{y} > c'Ay^*$ then $(\tilde{x}, \tilde{y})$ is not an optimal solution. This means that all optimal solutions verify $mx^* \geq Ay^*$.

$\square$

Remark 3. *There always exists an optimal solution of problem (2.2.1) that holds property (2.1.2).*

Proof. With the previous remarks we know that all feasible solutions hold the left-to-right implication. Let (x, y) be an optimal solution. If (x, y) holds property (2.1.2) we've finished.

Otherwise, there must exist some i such that doesn't hold the right-to-left implication. This means that

$$x_i = 0, \exists j \text{ s.t. } s_i \in C_j, y_j = 1$$

so

$$[Ay]_i \geq 1 > x_i = 0$$

If $i \in G$ repeating what was done in $\Leftarrow$ in the proof of remark 1 we reach a contradiction (substituting c' for $c'M$, which in this case because $i \in G$, $[c'M]_i = c_i > 0$ so the proof holds).

This means that necessarily $i \notin G$.

Defining $\tilde{x} = x + e_i$ then $[Ay]_i \geq 1 = \tilde{x}_i$ so $(\tilde{x}, y)$ is a feasible solution. Lastly because $i \notin G$, $[c'M]_i = 0$ so

$$c'M\tilde{x} = c'M(x + e_i) = c'Mx + c'Me_i = c'Mx$$

Now because $(\tilde{x}, y)$ is a feasible solution with the same objective value as an optimal solution, it also is an optimal solution.

After applying this process it may happen that there is some other i' for which $(\tilde{x}, y)$ does not hold property (2.1.2). It is enough to repeat the process until this is not the case. The result will be an optimal solution that does hold said property. $\square$

Remark 4. *There always exists an optimal solution of problem (2.2.2) that holds property (2.1.2).*

Proof. This proof follows the same scheme as the one proceeding it.

Again with the previous remarks we know that all feasible solutions hold the left-to-right implication. Let (x, y) be an optimal solution. If (x, y) holds property (2.1.2) we've finished.

Otherwise, there must exist some i such that doesn't hold the right-to-left implication. This means that

$$x_i = 0, \exists j \text{ s.t. } s_i \in C_j, y_j = 1$$

so

$$[Ay]_i \geq 1 > x_i = 0$$

If $i \in G$ repeating what was done in $\Leftarrow$ in the proof of remark 2 we reach a contradiction (substituting $c'x$ for $c'Mx$, which in this case because $i \in G$, $[c'M]_i = c_i > 0$ so the proof holds).

This means that necessarily $i \notin G$.

Defining $\tilde{x} = x + e_i$ then $[Ay]_i \geq 1 = \tilde{x}_i$. Also because $i \notin G$, $[c'M]_i = 0$ so

$$c'M\tilde{x} = c'M(x + e_i) = c'Mx + c'Me_i = c'Mx$$

This implies that $(\tilde{x}, y)$ is a feasible solution. It has the same objective value as an optimal solution because y remains the same, so it also is an optimal solution.

After applying this process it may happen that there is some other i' for which $(\tilde{x}, y)$ does not hold property (2.1.2). It is enough to repeat the process until this is not the case. The result will be an optimal solution that does hold said property. $\square$